VHDL NOT Gate Simulation Report

Logisim-evolution / Questa VHDL Simulation

Assignment	VHDL NOT Gate
Tool	Logisim-evolution with Questa
Objective	Verify that the VHDL NOT gate produces the inverse of input a.

1. VHDL Source Code

```
LIBRARY ieee;
USE ieee.std_logic_1164.all;

ENTITY not_gate IS
PORT (
a : IN std_logic;
y : OUT std_logic
);
END not_gate;

ARCHITECTURE behavior OF not_gate IS
BEGIN
y <= NOT a;
END behavior;
```

The statement `y <= NOT a;` assigns the logical inverse of input `a` to output `y`.

2. Logisim-evolution Circuit and Simulation

The VHDL entity was placed in the Logisim-evolution circuit and connected to one input pin (a) and one output pin (y). VHDL simulation was enabled through Questa.

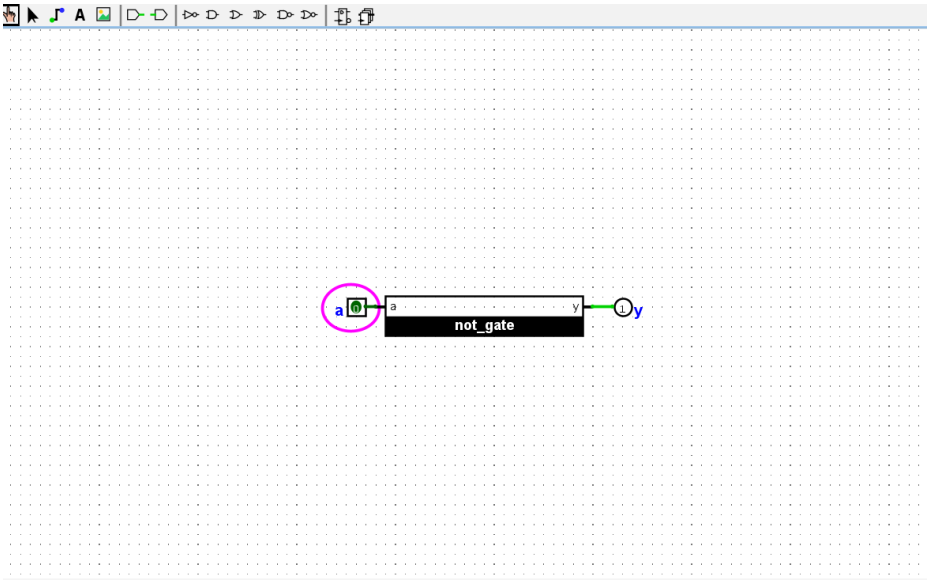


Figure 1. Simulation result for a = 0: the output is y = 1.

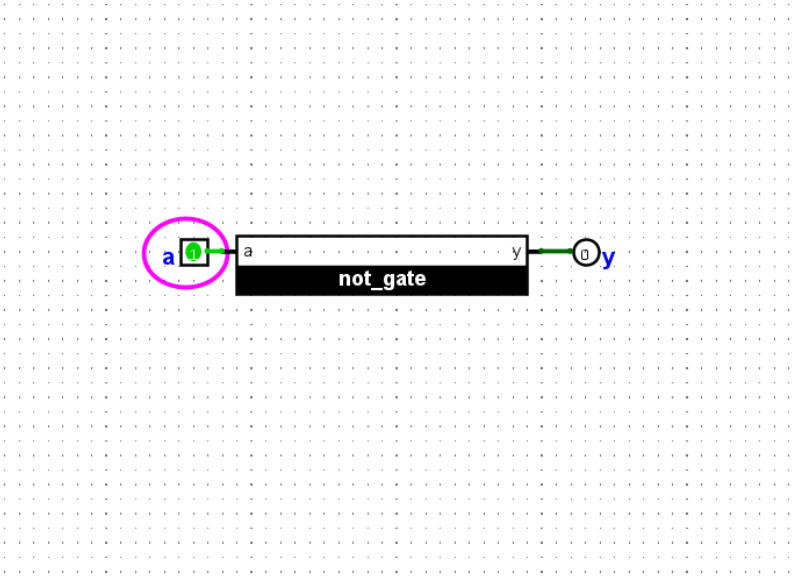


Figure 2. Simulation result for a = 1: the output is y = 0.

3. Results

Input a	Expected y	Observed y	Result
0	1	1	PASS
1	0	0	PASS

4. Conclusion

The VHDL NOT gate operated correctly in Logisim-evolution. When a = 0, y became 1, and when a = 1, y became 0. Therefore, the simulation results match the expected truth table of a NOT gate.